MICROSOFT:
Online test- Platform: Codility (hidden test cases with only one submission allowed per question) Duration: 90 mins A. Find least number greater than n with no two same adjacent digits. B. Find largest side of sqaure that can be formed on binary grid such that atleast two non overlapping squares of the required largest side can be formed.

The online coding round had 2 questions (Tree traversal and Graph) and this was conducted on Codility

OT Experience(Round 1) 18/08/2021

Duration : 95 mins
Number of coding questions : 2

Q1. Given a tree with N nodes (0 to N-1) and each node contains a letter either ‘a’ or ‘b’. We are given array A[ ] of length N where A[i] denotes the parent of i-th node. Node 0 is the root node with parent value = -1. Letters in nodes are given in the form of string s, where s[k] represents the letter in k-th node.
Find the number of vertices on the longest path in the tree such that no pair of adjacent vertices on the path would contain the same letter.
Constraints :
1 <= length of string <= 40,000

Input : s = “abbab” , A = [-1,0,0,0,2]
Output : 3 (longest path is 1->0->2)
Difficulty : Hard

Q2. We are given a string ‘s’ of length ‘n’ containing letters ‘a’ and ‘b’ only but also with some wildcard ‘?’ at any position [i : 0 to n-1]. So our task is to replace that ‘?’ with either ‘a’ or ‘b’ such that no 3 adjacent characters in ‘s’ are the same, i.e., neither “aaa” nor “bbb” should occur in the resulting string. Return the resulting string after replacing ? with suitable characters(a or b).
NOTE : It is guaranteed that a solution does exist & for multiple answers return any 1 valid string.
Range of n = [1...500,000]
Eg. s = “?aabb?”
Output = “baabba”
Eg. s = “?babba?bb?”
Output = “bbabbaabba”

Difficulty : Medium

Q1: There are N patients (numbered from 0 to n — 1) who want to visit the doctor. The doctor has S possible appointment slots, numbered from 1 to S. Each of the patients has two preferences. Patient K would like to visit the doctor during either slot A[k] or slot B[k]. The doctor can treat only one patient during each slot.

Is it possible to assign every patient to one of their preferred slots so that there will be at most one patient assigned to each slot?

Input: Two arrays A and B, both of N integers, and an integer S

1 <= N <= 10^6

1 <= S <= 2*10^6

1 <= A[i] <= S

1 <= B[i] <= S

no patient has two preference for the same slot i.e. A[i] != B[i]

Output: print "true" if it is possible to assign every patient to one of their preferred slots, one patient to one slot, and "false" otherwise.

Q2: Given an m x n binary matrix filled with 0's and 1's, find the two similar size non overlapping square of maximum number of cell containing only 1's and return maximum no of cell.

```
Input:   F T F T
         T T T F
         T T T T
         F T T T
Output: 4
```

Q1.A car manufacturer has data about the production processes of N different cars(numbered from 0 to N-1) and want to maximize the number of cars produced in the upcoming month.the manufacturing information is stored in array H,H[K] denote number of hours required to produce the Kth car.

There are two assembly lines in the factory,first one work for X and second one Y hours in a month.Every car can be constructed using either one of the lines.only one car can be constructed at a time.it is not possible to switch lines after starting production.what is the maximum no of different car that can be produced in the upcoming month?

input:H=[1,1,3],X=1,Y=1
output=2

Input: H={6,5,5,4,3} X=8,Y=9
Output=4
The cars that need 3 and 5 hour in first line
5 and 4 in second line

Q2.Our task concerns a city with N roads connecting N+1 junctions numbered from 0 to N. Roads connect junctions in such a way that each distinct pair of junctions is connected either by a direct road or through a path consisting of direct roads. There is exactly one way to reach any junction from any other junction.

There is an office next to junction 0 and a house next to every other junction. One person lives in each house. All the citizens work in the office and use a car to travel from their houses to the office. Each car has five seats and consumes 1 liter of fuel to move by one road (from one junction to a neighboring junction)

The citizens want to set up a carpool scheme that allows them all to travel to the office in a way that minimizes the total consumption of fuel. People can change cars at each house and then continue to travel together Calculate the fuel consumption in the optimal scheme.

The network of cities is described using arrays A and B, each of length N. Each pair A[K] B[K] for 0 SK<N specifies a road connecting junctions number A[K] and B[K]
Example

1. Given A = [0, 1, 1) and B-[1, 2, 31, there are N=3 houses, as in the image below. The function should return 3, as the optimal carpool scheme uses 3 liters of fuel

. Citizen 2 goes in his car from his house to house 1.
. Citizen 3 goes in her car from her house to house 1. .
 Citizens 2 and 3 go with citizen 1 in his car to the office.

2. Given A = [1, 1. 1.9.9.9,9, 7, 8) and B-12, 0, 3, 1, 6, 5, 4, 0, 0], there are N=9 houses, as in the image below. The function should retum 10, as the following carpool scheme is optimal:

. Citizens 7 and 8 go directly to the office in their cars (using 2 liters of fuel). .

Citizens 4, 5, and 6 go in their cars to house 9 (they use 3 liters of fuel) and then they go with citizen in her car to house 1 (for another 1 liter).9
Citizens 2 and 3 go in their cars to house 1 (for 2 liters of fuel), .
Now 7 citizens are at house 1, so they need to use at least two cars to move to the office (for another 2 liters of fuel)
Difficulty :Hard
Constrain: .N is an integer within the range [1..100,000);
each element of arrays A, B is an integer within the range [0..N]: . there is exactly one (possibly indirect) connection between any two distinct junctions.

Q1. Given a number, return the least number greater than this which doesn't have the same digits on two consecutive positions.
EX:
i/p : 64
o/p: 65

i/p:98
o/p:101 (99,100 both have same digits in two consecutive positions so 101 should be returned)

i/p:44221
o/p:45010

Q2. Given a binary matrix and an integer K with 1 as location of bird zone and 0 as freezone we need to find the number of locations in which the windmills can be installed.If a location say m[2][2] is bird zoned then all the cells which are in K distance from this cell are not suitable to install windmills.
The distance is defined as follows consider two nodes n1=m[i1][j1], n2=m[i2][j2]
Now distance(n1 , n2) = | i1-i2 |+| j1-j2 |
The distance between a node and the Bird zone (1 is bird zone in matrix) must be <=k
Ex:
Consider a matrix, consider the K value as 1,
1 0 0
0 0 0
0 0 1
0 1 0
O/p: 3
Explanation: All the nodes which are highlighted are not suitable or wind mills so the rest are useful and the count is 3.
1 0 0
0 0 0
0 0 1
0 1 0

Online Test:
2 coding questions(Easy-Medium level)
Duration: 95 Minutes
Question 1 : A Graph based question. Given a grid, the location of some birds on that grid and an integer k. A bird will not allow a factory to be built in any cell which is atmost k distance apart from it(distance=row dist+column dist). Find the number of cells where factories can be made.
I used the dfs approach for this.
Question 2 : Given an array of integers and an integer R. What is the maximum number of unique elements we can get by removing any possible sequence of R consecutive integers.
O(n^2) solution was getting partially accepted. So I used 2 maps to solve the question in linear time complexity.

Walmarts:

Max score: 50.00

Chaotic Subarrays

A subarray is said to be chaotic if the sum of the divisors count of the elements in the subarray is Odd.

Given an array A of N integers. Find the number of subarrays which are choatic in given array.

Example

If A = [4, 3, 9, 17] . Then number of chaotic subarrays is 6.

- The divisor count of A[1] : 4 is 3, because it has 3 divisors i.e.1, 2, 4.
- The divisor count of A[2] : 3 is 2, because it has 2 divisors i.e. 1 3.
- Similarly divisor count for 9 is 3 and 17 is 2.

6 chaotic subarrays are [4], [4,3], [9], [9,17], [3,9], [3,9,17].

Function Description

Complete the countSubarray function provided in the editor. This function takes the following 2 parameters and re[turn] number of chaotic subarrays.

- N : Represents the number of elements in array
- A : Represents the elements of the array

Input Format:

1. First line contain integer N, denoting number of elements in array.
2. Next line contain N space separated integers denoting the elements of array.

Constraints:

$1 \le N \le 10^5$

$1 \le A_i \le 10^{18}$

Output Format:

Output a single integer which is the number of choatic subarrays.

Sample input 1

```
4
12 36 2 21
```

Sample output 1

```
6
```

Same strings

Max. score 2000

You are given two strings s and t. Both the strings consist of lowercase English alphabets. You can perform the following three operations for multiple numbers of times:

- Operation 1: Select any two indexes i and j in string s and swap the characters of the selected indexes.
- Operation 2: Select any two indexes i and j in string t and swap the characters of the selected indexes.
- Operation 3: Select an index i in string s and select an index j in string t. Swap the characters of the selected indexes from both the strings.

You have to answer if it is possible to make the two strings equal. If it is possible, then print **YES**. Otherwise, print **NO**.

Input format

- First line: T denoting the number of test cases
- For each test case:
    - First line: s
    - Second line: t

$1 \leq T \leq 100$

$1 \leq size(s), size(t) \leq 10^5$

Sample Input 1

Copy

```
3
abc
def
abc
bca
abcdbc
adxxyy
```

Sample output 1

Copy

```
NO
YES
YES
```

Explanation

For the above example

... "def" we can't make both strings equal by using three operations.

... "bac"-> "abc". This is one of the possible way

Explanation

For the above example

Testcase 1: "abc" and "def" , we can't make both strings equal by using three operations.

Teastcase 2: "abc" and "bca" , if we swap 'c' and 'a' and then 'b' and 'a' from "bca" -> "bac" -> "abc". This is one of the possible way

Testcase 3: Similarly we can make s = abcdxy and t= abcdxy. This is one of the possible way.

Note: Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. There... your code must pass these hidden test cases to solve the problem statement.

Time Limit: 5.0 sec(s) for each input file

Memory Limit: 256 MB

Source Limit: 1024 KB

Marking Scheme: Score is assigned if any testcase passes

Allowed Languages: Bash, C, C++, C++14, C++17, Clojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java, Java 8, Java 14, JavaScript(Rhino), JavaScript(No... Kotlin, Lisp, Lisp (SBCL), Lua, Objective-C, OCaml, Octave, Pascal, Perl, PHP, Python, Python 3, Python 3.8, Racket, Ruby, Rust, Scala, Swift-4.1, Swift, T... Basic

New Submission

All Submissions